

Module A: Conjugate Directions & CG Intuition (Quadratic $Q \succ 0$)

We focus on the quadratic

$$f(x) = \frac{1}{2}x^\top Qx - b^\top x, \quad Q = Q^\top \succ 0, \quad g(x) = \nabla f(x) = Qx - b.$$

Conjugate directions (key theorem). Directions $d^{(0)}, \dots, d^{(m)}$ are Q -conjugate if

$$d^{(i)\top} Q d^{(j)} = 0 \quad (i \neq j).$$

- **Linear independence:** if $Q \succ 0$ and $\{d^{(i)}\}$ are Q -conjugate with $d^{(i)} \neq 0$, then they are linearly independent.
- **Why we care:** when you minimize a quadratic along a new Q -conjugate direction, you *don't undo* progress made along previous directions.

Subspace-optimality intuition (“CG optimizes one growing subspace at a time”). With exact line search along $d^{(0)}, d^{(1)}, \dots$, after k directions you are at the best point in

$$x^{(0)} + \text{span}\{d^{(0)}, \dots, d^{(k-1)}\}.$$

Equivalently, the gradient becomes orthogonal to that whole subspace:

$$g^{(k)\top} d^{(i)} = 0 \quad \text{for } i = 0, \dots, k-1.$$

Consequence: in exact arithmetic, CG reaches x^* in at most n steps (for n variables).

Module A.1: Conjugate Gradient Algorithm (what each step is doing)

CG maintains: iterate $x^{(k)}$, gradient $g^{(k)} = Qx^{(k)} - b$ (a.k.a. residual), and Q -conjugate search direction $d^{(k)}$.

1. **Init:** choose $x^{(0)}$.

2. **Residual:** $g^{(0)} = \nabla f(x^{(0)})$.

- *Purpose:* tells you how far you are from optimality ($g = 0$ iff optimal).

3. **First direction:** $d^{(0)} = -g^{(0)}$.

- *Purpose:* steepest descent start (cheap, always descent for SPD Q).

4. **Exact line search:**

$$\alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} + \alpha d^{(k)}) = -\frac{g^{(k)\top} d^{(k)}}{d^{(k)\top} Q d^{(k)}}.$$

- *Purpose:* minimize along current direction $\Rightarrow$ new gradient orthogonal to $d^{(k)}$.

5. **Update iterate:** $x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)}$.

6. **Update residual:** $g^{(k+1)} = \nabla f(x^{(k+1)})$.

- *Purpose:* stopping check; also explore the un-optimized space.

7. **Choose β_k to enforce conjugacy:**

$$\beta_k = \frac{g^{(k+1)\top} Q d^{(k)}}{d^{(k)\top} Q d^{(k)}} = \frac{\langle g^{(k+1)}, d^{(k)} \rangle_Q}{\|d^{(k)}\|_Q^2}$$

- *Purpose:* makes $d^{(k+1)} \perp_Q d^{(k)}$ (and in fact $\perp_Q$ to all previous); or equivalently, take out the $d^{(k)}$ component from $g^{(k+1)}$

8. **New direction:** $d^{(k+1)} = -g^{(k+1)} + \beta_k d^{(k)}$.

- *Purpose:* “corrected”/“orthogonalized” descent direction so new line search optimizes a *new* dimension.

🔗 Practice: Textbook 10.3 (one-line identity inside CG)

Let $f(x) = \frac{1}{2}x^\top Qx - x^\top b$ with $Q \succ 0$. In CG, directions satisfy

$$d^{(k)} = -g^{(k)} + \beta_{k-1}d^{(k-1)}, \quad d^{(k-1)\top} Q d^{(k)} = 0.$$

Show that

$$d^{(k)\top} Q d^{(k)} = -d^{(k)\top} Q g^{(k)}.$$

Answer: Use the update and Q -conjugacy with the previous direction:

$$d^{(k)\top} Q d^{(k)} = (-g^{(k)} + \beta_{k-1}d^{(k-1)})^\top Q d^{(k)} = -g^{(k)\top} Q d^{(k)} + \beta_{k-1} \underbrace{d^{(k-1)\top} Q d^{(k)}}_0.$$

Thus $d^{(k)\top} Q d^{(k)} = -g^{(k)\top} Q d^{(k)} = -d^{(k)\top} Q g^{(k)}$ (since $Q^\top = Q$).

🔗 Module B: Armijo Line Search (Backtracking)

Pick a descent direction d_k with $g_k^\top d_k < 0$ and choose α_k so that

$$f(x^{(k)} + \alpha_k d_k) \leq f(x^{(k)}) + c \alpha_k g_k^\top d_k, \quad c \in (0, 1).$$

- Start $\alpha = 1$; while Armijo fails, set $\alpha \leftarrow \beta \alpha$ ($\beta \in (0, 1)$, e.g. $\frac{1}{2}$).
- Terminates since $f(x^{(k)} + \alpha d_k) = f(x^{(k)}) + \alpha g_k^\top d_k + O(\alpha^2)$ and $g_k^\top d_k < 0$.

🔗 Practice: Sample Midterm Review: Nonlinear LS (Gauss–Newton + Armijo)

Consider minimizing

$$f(x) = \frac{1}{2} \sum_{i=1}^3 r_i(x)^2 \quad \text{where} \quad x = (A, \omega, \phi) \in \mathbb{R}^3, \quad r_i : \mathbb{R}^3 \rightarrow \mathbb{R}$$

satisfies

$$r_i(x) = y_i - A \sin(\omega t_i + \phi),$$

and y_i, t_i are real numbers.

Tasks:

- **Gauss–Newton:** Describe the Gauss–Newton method for solving this problem (write the Jacobian $J(x)$ explicitly and the linear system used to compute the step).
- **When is GN close to Newton?** (State a clear condition/intuition)
- **Armijo line search:** If the full step size $\alpha = 1$ does not decrease f , describe the backtracking line search with the Armijo condition for selecting α .

Answer: Gauss–Newton setup. Let $r(x) = [r_1(x), r_2(x), r_3(x)]^\top$ and $J(x) \in \mathbb{R}^{3 \times 3}$ with rows $\nabla r_i(x)^\top$. For each i ,

$$\frac{\partial r_i}{\partial A} = -\sin(\omega t_i + \phi), \quad \frac{\partial r_i}{\partial \omega} = -A t_i \cos(\omega t_i + \phi), \quad \frac{\partial r_i}{\partial \phi} = -A \cos(\omega t_i + \phi).$$

At iterate $x^{(k)}$, compute $p^{(k)}$ by solving the normal equations

$$(J(x^{(k)})^\top J(x^{(k)})) p^{(k)} = -J(x^{(k)})^\top r(x^{(k)}),$$

then set $x^{(k+1)} = x^{(k)} + \alpha_k p^{(k)}$ (often $\alpha_k = 1$ if accepted).

When GN is close to Newton. For nonlinear least squares,

$$\nabla f = J^\top r, \quad \nabla^2 f = J^\top J + \sum_{i=1}^3 r_i(x) \nabla^2 r_i(x).$$

GN uses $J^\top J$ as an approximation to $\nabla^2 f$ and drops the second term, so it is close to Newton when $\sum_i r_i(x) \nabla^2 r_i(x)$ is small—typically when the residuals $r_i(x)$ are small near a good fit / near the minimizer.

Armijo backtracking (if $\alpha = 1$ fails). Choose $c \in (0, 1)$ and $\tau \in (0, 1)$ (e.g. $c = 10^{-4}$, $\tau = \frac{1}{2}$). Start $\alpha = 1$ and shrink $\alpha \leftarrow \tau \alpha$ until

$$f(x^{(k)} + \alpha p^{(k)}) \leq f(x^{(k)}) + c \alpha \nabla f(x^{(k)})^\top p^{(k)}.$$

Then take $x^{(k+1)} = x^{(k)} + \alpha p^{(k)}$.